

Measuring Performance: Bias, Variance, and the Generalization Gap

Understanding Deep Learning

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MAI 604: Deep Learning

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Lecture Outline

- 1 Training vs. Generalization
- 2 Sources of Error
- 3 Mathematical Formulation
- 4 Reducing Error & The Trade-off
- 5 Double Descent
- 6 Model Selection & Validation

- A neural network can easily achieve perfect performance on the training data.
- Does it translate to unseen data?
- To understand real-world efficacy we must measure performance on a separate test set.
- Test error is primarily driven by three distinct components:
 - ① The inherent uncertainty in the task (Noise).
 - ② The amount of training data (Variance).
 - ③ The choice and flexibility of the model (Bias).

Experimental Setup: MNIST-1D

To explore model performance a1D version of the MNIST dataset.

Network Architecture:

- **Inputs:** $D_i = 40$ dimensions.
- **Hidden Layers:** Two hidden layers with $D = 100$ hidden units each.
- **Outputs:** $D_o = 10$ outputs passed through a softmax function to produce class probabilities.

Training Details:

- **Loss Function:** Multiclass cross-entropy loss.
- **Optimizer:** Stochastic Gradient Descent.
- **Hyperparameters:** Batch size 100, Learning rate 0.1.
- **Duration:** Trained for 150 epochs.

Results: Training vs. Test Performance

Observations on MNIST-1D:

- Training error rapidly decreases to zero.
- Test error plateaus at around 40%. The model **fails to generalize well**.
- **Notice the Loss:** While test error remains constant, test loss *increases* after a certain point.

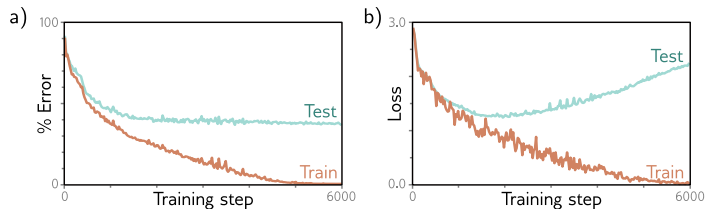


Fig 1: Training vs. Test Error and Loss

Understanding the Ground Truth

To visualize errors, consider a 1D least-squares regression task.

- **Ground Truth:** The true underlying function is quasi-sinusoidal (solid black line).
- **Data Generation:** We sample inputs uniformly across $[0, 1]$ and add **Gaussian noise**.
- The gray region represents ± 2 standard deviations of the noise.

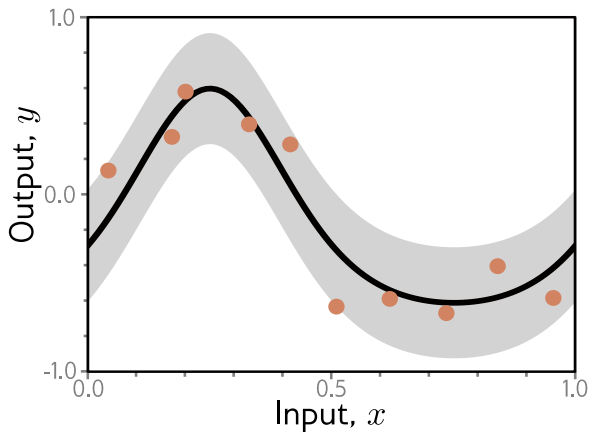


Fig 2: 1D Regression problem and noisy samples

Noise, Bias, and Variance

When a model fails to generalize perfectly, the error stems from three sources:

Noise

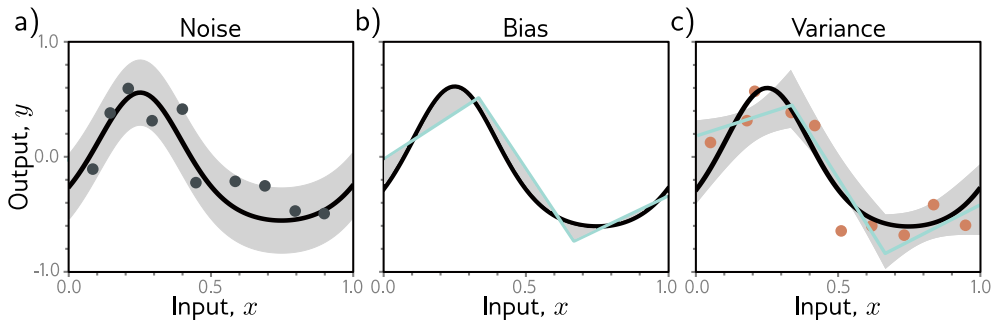
Inherent stochasticity in the data (missing variables, measurement error). **Insurmountable**.

Bias

Model inflexibility. The model cannot perfectly fit the true underlying function, even with optimal parameters.

Variance

Variability in the fitted model caused by the idiosyncrasies of a limited, noisy training dataset.



Standard Mathematical Formulation of Test Error

Let's derive the standard **Bias-Variance Decomposition** for squared error.

Assume:

$$y = f(x) + \epsilon \quad \text{where} \quad \mathbb{E}[\epsilon] = 0, \quad \text{Var}(\epsilon) = \sigma^2$$

Let $\hat{f}(x; D)$ be our trained model based on a dataset D . We want to find the expected test error at a fixed test point x , taken over all possible training datasets D :

$$\text{Expected Error} = \mathbb{E}_D \left[(y - \hat{f})^2 \right]$$

Bias-Variance Decomposition (Derivation)

We can expand the expected squared error:

$$\begin{aligned}\mathbb{E}_D \left[(y - \hat{f})^2 \right] &= \mathbb{E}_D \left[(f(x) + \epsilon - \hat{f})^2 \right] \\ &= \mathbb{E}_D \left[(f(x) - \hat{f})^2 + \epsilon^2 + 2\epsilon(f(x) - \hat{f}) \right]\end{aligned}$$

Since ϵ has zero mean and is independent of $\hat{f}$, the cross term vanishes, and $\mathbb{E}[\epsilon^2] = \sigma^2$:

$$= \mathbb{E}_D \left[(f(x) - \hat{f})^2 \right] + \sigma^2$$

Now, add and subtract the expected model prediction $\mathbb{E}_D[\hat{f}]$ inside the squared term:

$$\begin{aligned}\mathbb{E}_D \left[(f(x) - \hat{f})^2 \right] &= \mathbb{E}_D \left[\left((f(x) - \mathbb{E}_D[\hat{f}]) + (\mathbb{E}_D[\hat{f}] - \hat{f}) \right)^2 \right] \\ &= (f(x) - \mathbb{E}_D[\hat{f}])^2 + \mathbb{E}_D \left[(\mathbb{E}_D[\hat{f}] - \hat{f})^2 \right] + \text{cross term}\end{aligned}$$

The expected value of the cross term is zero!

The Fundamental Trade-off Equation

Putting it all together, we arrive at the classic decomposition:

Bias-Variance Decomposition

$$\mathbb{E} \left[(y - \hat{f})^2 \right] = \underbrace{(f(x) - \mathbb{E}[\hat{f}])^2}_{\text{Bias}^2} + \underbrace{\mathbb{E} \left[(\hat{f} - \mathbb{E}[\hat{f}])^2 \right]}_{\text{Variance}} + \underbrace{\sigma^2}_{\text{Noise}}$$

- **Bias²**: How far off the average model is from the true function.
- **Variance**: How much the model predictions fluctuate based on the specific training data sampled.
- **Noise**: The irreducible error σ^2 inherent to the problem.

Choosing Between Estimators

Scenario

We want to estimate the mean age of a population. Let's assume the true mean is $\mu = 40$ and the variance is $\sigma = 12$. Which estimator is better?

- **Estimator 1:** Unbiased sample mean from $n = 10$ people.

$$\hat{\theta}_1 = \bar{X}_{10}$$

- **Estimator 2:** A slightly biased estimator from a huge sample of $n = 1000$ people.

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Let's compare them using MSE:

- $\text{MSE}(\hat{\theta}_1) = \text{Var}(\hat{\theta}_1) + \text{Bias}(\hat{\theta}_1)^2 = \frac{\sigma^2}{10} + 0^2 = \frac{144}{10} = 14.4.$

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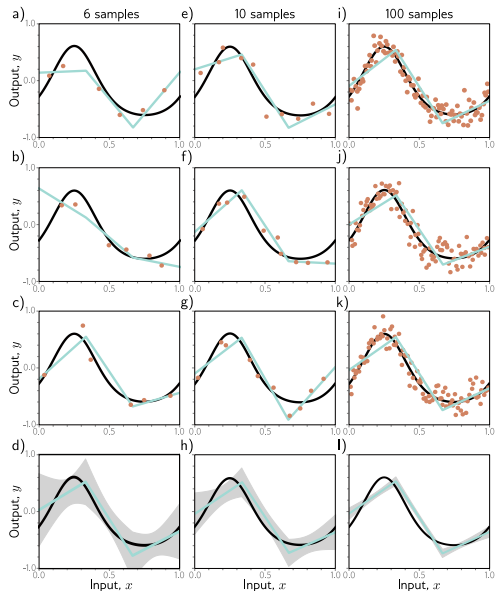
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- $\text{MSE}(\hat{\theta}_2) = \text{Var}(\hat{\theta}_2) + \text{Bias}(\hat{\theta}_2)^2 = \frac{\sigma^2}{1000} + (0.1)^2 = 0.154.$

Reducing Variance

How to reduce variance?

- Variance measures how much the model fluctuates when trained on different datasets D . This is associated with *overfitting*.
- We can reduce variance by increasing the quantity of training data.
- Other remedies include regularization and early stopping.



How to reduce bias?

- Bias results from the model's inability to describe the true function, often due to *underfitting*.
- We reduce bias by making the model more flexible.
- In neural networks, this means adding more hidden units or layers.

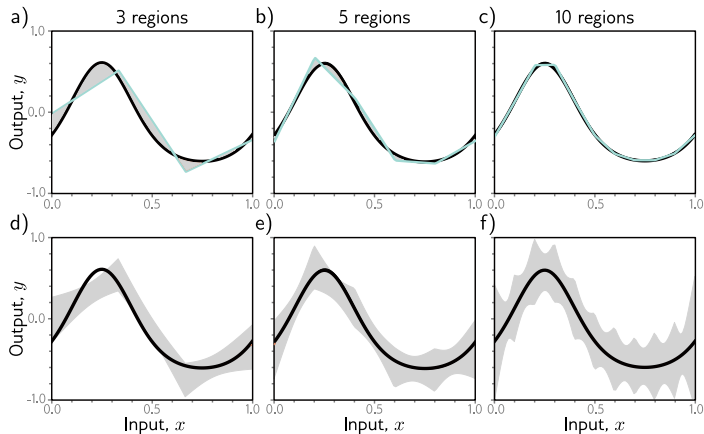


Fig 5: Increasing capacity (linear regions) reduces bias (a-c) but increases variance (d-f).

The Danger of Capacity: Overfitting

Overfitting:

- A model with too much capacity for a fixed-size dataset doesn't necessarily produce better predictions.
- It dedicates its flexibility to modeling the noise.
- The variance increases massively.

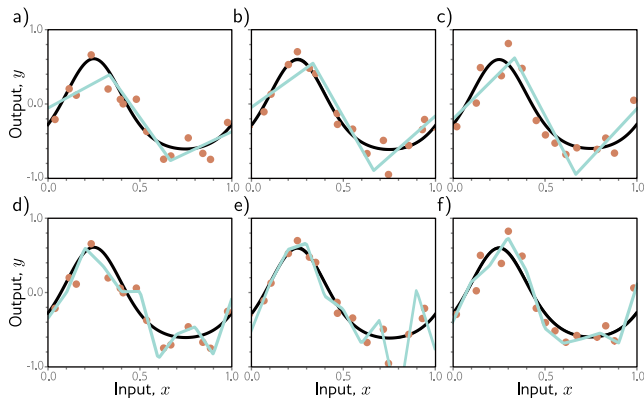


Fig 6: Overfitting: differences between fitted curves are larger when 10 regions (bottom) vs 3 regions (top).

The Bias-Variance Trade-off

Classical View

- As model capacity increases, bias **decreases**.
- As model capacity increases, variance **increases**.
- The total expected error is the sum of these terms.
- Goal: Find the optimal capacity (the "sweet spot") that minimizes the sum of bias and variance.

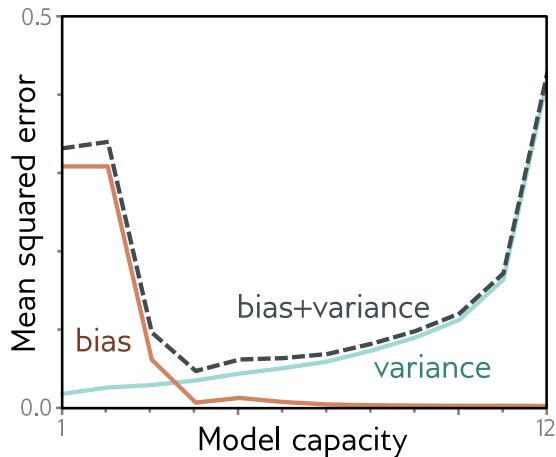


Fig 7: Expected error forms a U-shape.

Modern Theory: Double Descent

- The bias-variance trade-off assumes error behaves like a U-shape.
- However, in modern deep learning, models are heavily over-parameterized i.e. they have more parameters than training data points.
- Once the model fits the training data perfectly (interpolation point), classical theory suggests variance should be infinitely bad.
- Surprisingly, as we continue adding capacity past the point of memorization, test error goes down again! This is called **Double Descent**.

Inductive Bias & Smooth Interpolation

Why does over-parameterization help?

- With just enough capacity to memorize data, the function contorts wildly.
- With excess capacity, the network's inductive bias prioritizes **smoother functions**.

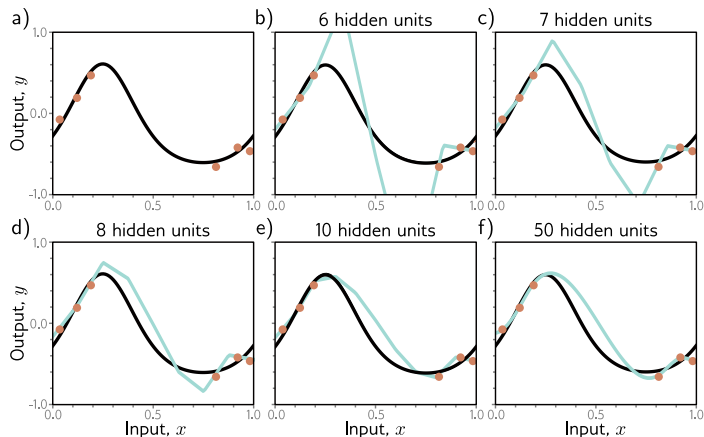


Fig 8: Increasing capacity allows for significantly smoother interpolation between points.

Parameters vs. Hyperparameters

- **Parameters:** Weights and biases. We learn these automatically using optimization algorithms like SGD.
- **Hyperparameters:** Structural and algorithmic settings chosen *before* training begins.
 - *Model capacity:* Number of hidden layers, number of units per layer.
 - *Learning algorithm:* Learning rate, batch size, choice of optimizer.
- **Hyperparameter Search:** Finding the best configuration is difficult. Spaces are large, often discrete, and non-differentiable. We must *empirically* evaluate multiple configurations.

The Need for a Validation Set

The Problem:

- If we evaluate multiple hyperparameter choices on the test set and pick the best one, we commit a data leakage error.

The Solution (Three-way Split):

- **Training Set:** Used strictly to fit the model parameters (weights).
- **Validation Set:** A proxy dataset used to evaluate performance and [select the best hyperparameters](#).
- **Test Set:** Kept strictly sequestered until the very end to provide a final, unbiased measure of true performance.

Cross-Validation (K -Fold)

What if we don't have enough data?

- Setting aside a large validation set reduces our training data.
- A single validation split might be noisy.
- **Cross-Validation** solves this by partitioning the training data into K equal subsets (folds).
- We train on $K - 1$ folds and validate on the 1 remaining fold. We repeat this K times and **average the validation performance**.

Note: While highly effective, K -Fold CV is computationally expensive. In deep learning, where training one model takes days, a single large validation set is often preferred if data allows.

5-Fold CV Example

Fold 1: [V] [T] [T] [T] [T]

Fold 2: [T][V] [T] [T] [T]

Fold 3: [T] [T][V] [T] [T]

Fold 4: [T] [T] [T] [V] [T]

Fold 5: [T] [T] [T] [T] [V]

$V = \text{Validation}, T = \text{Train}$

- **Test Error vs. Training Error:** Good training performance doesn't guarantee generalization. Use a test set.
- **Three Sources of Error:**
 - **Noise:** Insurmountable data stochasticity.
 - **Bias:** Systemic inflexibility in the model.
 - **Variance:** Fluctuation caused by limited training data.
- **Standard Math:** $(y - \hat{f})^2 = \text{Bias}^2 + \text{Variance} + \text{Noise}$.
- **Modern Regime:** Over-parameterized models undergo **Double Descent**, leveraging inductive biases to find smooth interpolations.
- **Model Selection:** Use a **Validation Set** (or **Cross-Validation**) to tune hyperparameters without leaking test data information.